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Abstract

This internal report presents the formulas for the response matrix and dispersion derivatives with respect to the quadrupole strengths. The equations validity is extended to the thick quadrupole case. The formulas hold only in the constant momentum case, the discrepancies with the constant path case are assessed for the ALBA lattice. These formulas make the optics correction, in particular the LOCO analysis, several times faster.

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References

[1] J. Safranek, *Experimental determination of storage ring optics using orbit response measurements*, Nucl. Inst. and Meth. A388 (1997) 27.

[2] Accelerator toolbox 2.0: <https://github.com/atcollab/at>

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[5] K. Brown, *A First and Second Order Matrix Theory for the Design of Beam Transport Systems and Charged Particle Spectrometers*, SLAC-75, june 1982.

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1 Introduction

Analytical formulas for the Jacobian of the Orbit Response Matrix (ORM) can provide a significant numerical improvement for LOCO calculations. In this report we detail the new Analytical Calculations on Orbit Response Matrices by AT (ACalORAT).

During this work analytical formulas for different ring parameters are derived, implemented and benchmarked in Python. Combined function dipoles (CFD) provide a new unique challenge when correcting for their strength as adjusting them.

2 ACalORAT

Acalorat is the ATcollab module created to perform analytical calculations on ring parameters.

2.1 Elements

In a calculation, the information on all elements of a set is stored in an instance of the Elements class. It contains a pointer to the ring, the indices of the elements in the set and all the relevant optical functions. Several methods are available to treat the optics of the elements for different calculations.

- **Correct entrance:** Optics in AT correspond to the optics at the entrance of an element, but for transfer matrices to be applied one has to consider the optics after the fringe quadrupole, which this function can be called to correct for.
- **Average:** Uses the entrance correction and Twiss transfer matrix to compute the average dispersion inside of an Element
- **broadcasters:** Takes as input broadcast Builds broadcasters for the NumPy tensor calculations. Takes all the public variables of the class (the ones not starting with _)

2.2 AnaORM

The AnaORM class contains all the information needed for a calculation in a given ring. It is started

To run a calculation as detailed in the auxiliary documents. To perform a calculation the ring must be imported into AT and a dictionary with the structure detailed in the documentation. Afterwards, the AnaORM class is initialized by passing the ring, the index dictionary and optionally some previously computed optics. A couple of short scripts are provided to read the lattice and indices for ALBA and ALBAII If no optics have been introduced, AnaORM.get_optics must be run to get the optics. Afterwards, the elements involved in the calculation have to be initialized by specifying the dimension they will occupy in the final result, for example, to calculate the derivative of dispersion with respect to a quadrupole, one should run:

```
import at

from ACalORAT import read
from ACalORAT import AnaORM
```

```

ring, ind = read.ALBAII(ROOT / "data" / "ring_a2.mat")

calc = AnaORM.AnaORM(ring,"h", ind)
calc.assign_optics()

calc.bpm.broadcasters(0, 2)
calc.CFD.broadcasters(1, 2)

dni_dqk = calc.dni_dqk_integral(calc.bpm, calc.CFD)

```

And to calculate the derivative of the dispersion with respect to changing the quadrupole component in pure quadrupoles, for example, one should just change CFD in the previous code for quad.

Most generic Calculations can be run in a straight forward way by the functions already included in ana_utils. Open the webpage, Google

3 FULL quadrupole response matrix

For thick quadrupoles, the derivative with respect to the ORM is already solved by Zeus by integrating the contribution on the derivative of the ORM Jacobian from thin quadrupoles, as seen in Fig. 2. In the horizontal direction, the error is due to the fact that the derivative of dispersion matches everything even better. We use entrance optics provided by AT and correct by the thin entrance quadrupole.

For thin elements, the ORM is, letting i denote bpm index and j correctors:

$$R_{ij} = \frac{\sqrt{\beta_i \beta_j}}{2 \sin(\pi \nu)} \cos(|\phi_i - \phi_j| - \pi \nu) \quad (1)$$

In the horizontal direction, the inherent fixed RF frequency imposed by the cavities in the ring adds an additional term to the response matrix. given by $\eta_i \langle \eta_j \rangle / (\alpha_c C)$. Where η_l denotes the dispersion in a given element, α_c the momentum compaction factor and C the circumference of the ring.

Taking the derivative of the horizontal term, one gets the Jacobian of the ORM with respect to quadrupoles the results in 2 are obtained.

$$\begin{aligned}
\frac{dR_{i,j}}{dq_k} &= \frac{\partial R_{i,j}}{\partial \beta_i} \frac{d\beta_i}{dq_k} + \frac{\partial R_{i,j}}{\partial \beta_j} \frac{d\beta_j}{dq_k} + \frac{\partial R_{i,j}}{\partial \nu} \frac{d\nu}{dq_k} + \frac{\partial R_{i,j}}{\partial \psi_i} \frac{d\psi_i}{dq_k} + \frac{\partial R_{i,j}}{\partial \psi_j} \frac{d\psi_j}{dq_k} \\
&= \mp \frac{\sqrt{\beta_i \beta_j} \beta_k L_k}{8 \sin(\pi \nu) \sin(2\pi \nu)} [C_{i,j,1} [C_{i,k,2} + C_{j,k,2} + 2 \cos^2(\pi \nu)] \\
&\quad + S_{i,j,1} [S_{i,k,2} - S_{j,k,2} + \sin(2\pi \nu) (2\theta(\psi_i - \psi_k) - 2\theta(\psi_j - \psi_k)) - s(\psi_i - \psi_j)]],
\end{aligned} \quad (2)$$

Where the sign is positive for the horizontal plane and negative for the vertical plane. ?? In Zeus report, this formula has been integrated to consider the effect of both thick correctors and thick quadrupoles, yielding 2

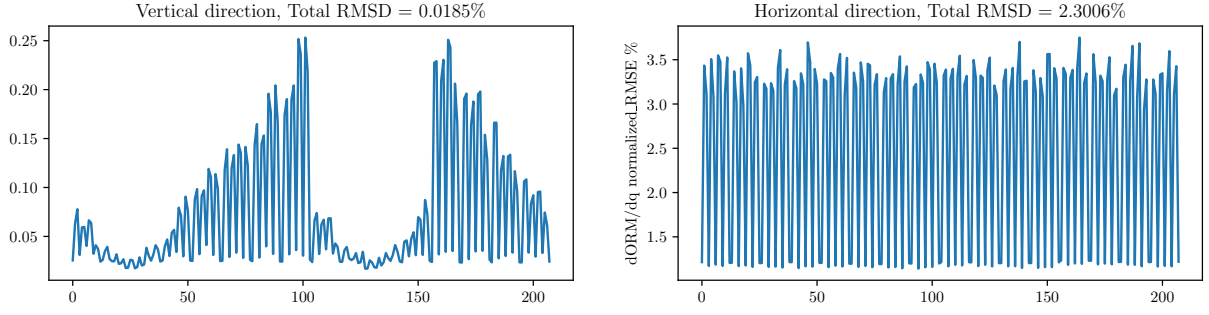


Figure 1: Caption

3.1 Dispersion derivative

To calculate the derivative of dispersion with respect to changing a quadrupole, we base ourselves on the fact that for a thin quadrupole k in a bpm i

$$\frac{d\eta_i}{dq_k} = \eta_k R_{ij} \quad (3)$$

Using the Twiss transfer matrix for dispersion then and considering the optics one can integrate to get the effect for a thick quadrupole. Using the convention that when there is no explicit s dependence we make reference to the optics at the entrance of the element. For a bending magnet with quadrupole strength k :

$$\eta(s) = \cos(\sqrt{k}s)\eta + \frac{\eta'}{\sqrt{k}}\sin(\sqrt{k}s) + \frac{h}{k}(1 - \cos(\sqrt{k}s)) \quad (4)$$

We can use equations derived by Zeus:

$$C_{ij1}(s) = C_{ij1} \cos(\psi_j(s) - \psi_j) + S_{ij1} \sin(\psi_j(s) - \psi_j) \quad (5)$$

We can rewrite this equations to invoke the integrals already solved by Zeus using the optics evolution inside of magnets:

$$\cos(\sqrt{k}s) = \frac{\sqrt{\beta(s)}}{\sqrt{\beta}} \cos(\psi(s) - \psi) + \frac{\alpha}{\sqrt{\beta}} \sqrt{\beta(s)} \sin(\psi(s) - \psi) \quad (6)$$

$$\sin(\sqrt{k}s) = \sqrt{k\beta} \sqrt{\beta(s)} \sin(\psi(s) - \psi) \quad (7)$$

Expanding Eq. ??? def for a CFD with length L .

$$\frac{d\eta_i}{dq_j} = \frac{1}{L} \int_0^L \left(\cos(\sqrt{k}s)\eta + \frac{\eta'}{\sqrt{k}}\sin(\sqrt{k}s) + \frac{h}{k}(1 - \cos(\sqrt{k}s)) \right) \frac{\sqrt{\beta_i\beta_j(s)}}{2\sin(\pi\nu)} \cos(|\psi_i - \psi_j(s)| - \pi\nu) ds \quad (8)$$

We can rewrite this equations to invoke the integrals already solved by Zeus using the optics evolution inside of magnets:

$$\cos(\sqrt{k}s) = \frac{\sqrt{\beta(s)}}{\sqrt{\beta}} \cos(\psi(s) - \psi) + \frac{\alpha}{\sqrt{\beta}} \sqrt{\beta(s)} \sin(\psi(s) - \psi) \quad (9)$$

Using the integrals and trigonometric identities

$$\begin{aligned}\int \beta_k(s) \cos^2(\psi_k(s) - \psi_k) ds &= (I_{k0} + I_{kc2})/2 \\ \int \beta_k(s) \sin^2(\psi_k(s) - \psi_k) ds &= (I_{k0} - I_{kc2})/2 \\ \int \beta_k(s) \sin(\psi_k(s) - \psi_k) \cos(\psi_k(s) - \psi_k) ds &= I_{ks2}/2\end{aligned}$$

Using all the formulas, the derivative of dispersion with respect to changing the quadrupole strength in a CFD is:

$$\begin{aligned}\frac{\partial \eta_i}{\partial q_k} = & -\frac{1}{2 \sin(\pi \nu)} \sqrt{\frac{\beta_i}{\beta_k}} \left[\eta_k \left(C_{ik} \frac{I_{k0} + I_{kc2}}{2} + \alpha_k S_{ik} \frac{I_{k0} - I_{kc2}}{2} + \frac{I_{ks2}}{2} (\alpha_k C_{ik} + S_{ik}) \right) \right. \\ & + \eta'_k \beta_k \left(C_{ik} \frac{I_{ks2}}{2} + S_{ik} \frac{I_{k0} - I_{kc2}}{2} \right) \\ & \left. + \frac{h_k}{K_k} \left(\beta_k L_k (I_{kc1} C_{ik} + I_{ks1} S_{ik}) - \left(C_{ik} \frac{I_{k0} + I_{kc2}}{2} + \alpha_k S_{ik} \frac{I_{k0} - I_{kc2}}{2} + \frac{I_{ks2}}{2} (\alpha_k C_{ik} + S_{ik}) \right) \right) \right] \quad (10)\end{aligned}$$

In section 4.1.1 this formula is validated for the ALBAII lattice.

With this term, the full derivative of the dispersion term in the horizontal response matrix can be analytically derived yielding: In a corrector, the average dispersion is

$$\langle \eta_i \rangle = \eta_i + \eta'_i L/2 \Rightarrow \frac{\langle \eta_i \rangle}{dq_k} = \frac{\eta_i}{dq_k} + \frac{\eta'_i}{dq_k} L/2 \quad (11)$$

As dispersion is linear in a drift and all corrector in ALBAII are preceded by a drift, the derivative of the dispersion derivative can be calculated by taking the difference.

Finally, the derivative of the term is, by virtue of the product rule:

$$d = \frac{\eta_i \frac{\langle \eta_j \rangle}{dq_k}}{\alpha_c C} + \frac{\langle \eta_j \rangle \frac{\eta_i}{dq_k}}{\alpha_c C} - \frac{\eta_i \langle \eta_j \rangle}{\alpha_c^2 C} \frac{d\alpha_c}{dq_k} \quad (12)$$

Dropping the mcf derivative term, we get, as seen in Fig. 2 a significant improvement in the Jacobian. With the remaining error being attributed to the lacking MCF derivative.

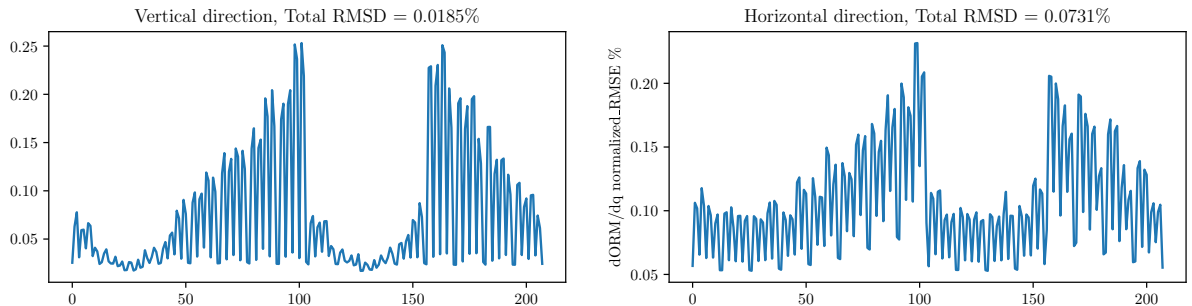


Figure 2: Analytical comparison of the ORM Jacobian with respect to changing quadrupoles

this improves significantly the error in the horizontal transverse dimension, the extra error left can be attributed to the derivative with respect to the momentum compaction factor on that

term, which could be obtained by integrating the square of the dispersion inside of the element using the twiss transfer matrix again.

4 Numerical calculations

On the other hand, recalling that dispersion is also a closed orbit, the derivative with respect to changing a thin quadrupole has an effect on the dispersion in the rest of the ring!

4.1 Validated quantities

4.1.1 Quadrupole dispersion derivative

The derivative of dispersion with respect to quadrupoles has been validated.

4.1.2 corrector activation

This naturally leads to two different relevant effects an energy change and a closed orbit distortion.

4.1.3 Energy change

For the energy change, when the active feedback, total corrector drift is avoided by imposing the condition:

$$\sum \langle \eta_i \rangle \theta_i = 0 \quad (13)$$

Thus, when changing these correctors

4.2 Closed orbit distortion in sextupoles

It is niceee

5 Work left to do

#A lot :(and):

In this section the author details by order of importance the work left to to.